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METHODS

A Mixed-Language Transformer Encoder Architecture for Social Media Phishing: Vietnamese-English Case Study

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ABSTRACT Social media phishing has become increasingly prevalent in regions where attackers exploit informal writing styles, multilingual expressions, and code-mixed compound structures. These characteristics pose significant challenges for detection systems built on monolingual or general-purpose multilingual models. In this paper, we propose MLTEA, a mixed-language dual-encoder architecture designed to model Vietnamese–English code-mixed compound structures commonly observed in social media phishing. The framework integrates parallel pretrained encoders, performs token-level fusion, and employs a gated reweighting mechanism to capture cross-lingual semantic dependencies. Experimental results show that MLTEA consistently outperforms monolingual, multilingual, and hybrid baselines, achieving an F_1 score of 0.94 under clean conditions while maintaining strong performance under structurally obfuscated inputs, demonstrating robust generalization under controlled compound-level and code-mixed perturbations.

INDEX TERMS Social media phishing, code-mixed compound structures, token-level fusion, transformer encoder.

I. INTRODUCTION

Social media phishing (SMP) has emerged as a critical security threat in modern digital communication environments. Unlike traditional phishing, SMP exploits social networking platforms and instant messaging applications, including private messages, group chats, public posts, and social comments, to deceive users into disclosing sensitive information. Contemporary platforms such as Facebook, Instagram, WhatsApp, and Zalo enable rich, real-time interactions through hyperlink sharing, multimedia content, and rapid information propagation. While these features enhance user engagement, they also significantly expand the attack surface, allowing phishing campaigns to propagate rapidly through trusted social connections.

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Mobile-based phishing attacks, including smishing, have become increasingly prevalent and pose significant security risks in real-world communication environments [1], [2]. According to the Anti-Phishing Working Group [3], phishing attacks are increasingly shifting from email to social and messaging platforms. Recent reports indicate that these campaigns redirect victims to counterfeit delivery portals, fraudulent banking interfaces, and fake account verification pages. This threat is particularly pronounced in Vietnam, where widespread Vietnamese–English code-mixed expressions introduce substantial linguistic variability and structural obfuscation, posing significant challenges for automated detection systems.

Recent advances in phishing detection have increasingly leveraged deep learning models to capture complex linguistic and structural patterns in malicious messages [4], [5]. These approaches benefit from representation learning capabilities that reduce reliance on handcrafted features and improve generalization across diverse phishing scenarios. In addition, hybrid frameworks that integrate textual content with URL-based features have demonstrated improved detection performance by exploiting complementary signals across modalities [6].

However, despite these advancements, existing methods largely rely on standard tokenization pipelines and assume well-formed linguistic inputs. As previously illustrated through compound token examples (e.g., `cap_nhat_password`), such assumptions break down when multiple semantic units are fused within a single surface form. In these cases, subword tokenization fragments semantically coherent units, disrupting the alignment between action cues and credential-related targets. Consequently, current models fail to preserve cross-lingual semantic interactions at the token level, limiting their effectiveness under compound-level obfuscation and mixed-language SMP scenarios. Attackers commonly exploit code-mixed expressions that embed security-related keywords (e.g., `password`, `verify`) within conversational Vietnamese text, often combined with obfuscated URLs [7]. In many cases, Vietnamese action phrases and English keywords are fused into compound tokens (e.g., `cap_nhat_password`), which obscure canonical word boundaries and weaken the visibility of phishing cues during tokenization. Similar mixed-language compound constructions also appear in URLs, such as `capnhatpassword-bank.vn` or `baomatOTPbank.vn`, where Vietnamese and English lexical units are concatenated into a single domain string, further masking semantic boundaries.

Such constructions introduce a fundamental challenge for tokenization-based models, as they encode multiple semantic units across languages within a single surface form. In the example `cap_nhat_password`, `cap_nhat` ("update") conveys an action intent, while `password` denotes a sensitive credential target; the phishing intent arises from their interaction rather than isolated tokens. However, standard subword tokenizers in multilingual Transformers fragment these tokens (e.g., `cap ##nh ##at ##pass ##word`), disrupting semantic alignment and weakening the association between action and target cues.

Existing multilingual Transformer models [8], [9], [10] rely on generic subword vocabularies that are not optimized for hybrid Vietnamese–English constructions. Consequently, they tend to over-fragment code-mixed compound tokens, which degrades semantic locality and weakens cross-lingual attention alignment, ultimately reducing discriminative power under structurally obfuscated SMP inputs.

In contrast, PhoBERT [11], trained on large-scale Vietnamese corpora, provides more linguistically coherent representations of Vietnamese morphology and segmentation. For the same token, PhoBERT is more likely to produce (e.g., `cap_nhat password`), preserving the Vietnamese compound structure while maintaining the integrity of the embedded English keyword. This improves semantic locality and strengthens the association between action phrases and credential-related cues. However, as a monolingual model, PhoBERT does not explicitly capture cross-lingual interactions, limiting its effectiveness in code-mixed SMP scenarios.

In such settings, **false negatives** represent the most critical failure mode, as malicious messages bypass detection and directly impact end users. This issue is particularly pronounced when phishing intent is embedded within structurally fused mixed-language tokens, where tokenization-induced semantic dilution hinders reliable classification. These limitations highlight the need for a unified architecture that jointly models language-specific structure and cross-lingual semantics.

To address this challenge, we propose a Mixed-Language Transformer Encoder Architecture (MLTEA) for SMP detection. The framework integrates a monolingual Vietnamese encoder (PhoBERT) and a multilingual encoder (XLM-RoBERTa) through token-level feature fusion, enabling the model to capture both language-specific morphological structure and cross-lingual semantic interactions. To further enhance contextual modeling, a Bidirectional Long Short-Term Memory (BiLSTM) layer is applied over the fused representations to introduce sequential inductive bias, improving robustness to irregular and structurally obfuscated text patterns. The main contributions of this paper are summarized as follows:

- We propose MLTEA that integrates PhoBERT and XLM-RoBERTa through token-level fusion to effectively model Vietnamese–English code-mixed compound structures,
- We demonstrate that incorporating a BiLSTM over fused representations improves recall for structurally obfuscated phishing messages by reinforcing local sequential dependencies,
- We design a lightweight decomposition-based preprocessing strategy that preserves code-mixed structures while improving compatibility with subword tokenizers,
- We construct a scenario-controlled SMP dataset and conduct extensive experiments showing that the proposed model outperforms monolingual, multilingual, and hybrid baselines in both accuracy and robustness.

The rest of this paper is organized as follows. Section II reviews the related literature on SMP detection and code-mixed processing. Section III introduces the proposed architecture. Section IV presents the experimental setup and analyzes the results.Section VI concludes the paper, discusses its limitations, and outlines future research directions.

II. RELATED WORK

This section is organized into three subsections: monolingual Transformer encoders, multilingual Transformer encoders, and Vietnamese–English SMP resources.

TABLE 1. Comparison of transformer encoder models in SMP detection.

Model	Type	Pretraining Corpus	Code-Mix Handling	Compound Token Handling	Limitation in SMP Context
BERT [9]	Monolingual (English)	English Wikipedia, BooksCorpus	Not supported	Not supported	English-only; Vietnamese tokens are poorly represented and code-mixed inputs are severely fragmented
PhoBERT [11]	Monolingual (Vietnamese)	Large-scale Vietnamese corpora	Not supported	Partial support	Preserves Vietnamese compound structure but cannot capture embedded English tokens or cross-lingual interactions
mBERT [10]	Multilingual	104 languages (Wikipedia)	Supported	Not supported	Shared vocabulary reduces language-specific precision; intra-token code-mixed disrupts subword alignment
XLM-RoBERTa [26]	Multilingual	100 languages (CommonCrawl)	Supported	Not supported	SentencePiece tokenisation fragments compound tokens; no explicit compound-boundary modeling
InfoXLM [27]	Multilingual	Multilingual, parallel corpora	Supported	Not supported	Contrastive alignment objective; lacks robustness to noisy and structurally obfuscated compound tokens
mT5 [28]	Multilingual	Large-scale multilingual text	Supported	Not supported	Sequence-to-sequence architecture; lacks fine-grained token-level discriminative capacity for classification
LaBSE [30]	Multilingual (sentence-level)	Parallel corpora (100+ languages)	Supported	Not supported	Sentence-level pooling discards token-level phishing cues embedded in compound tokens
Sentence-BERT [32]	Multilingual (sentence-level)	Sentence pairs	Supported	Not supported	Optimised for semantic similarity; insufficient sensitivity to local token-level credential and URL cues
CAEoT [25]	Monolingual (English)	Task-specific corpora	Not supported	Supported	Explicit compound decomposition but monolingual scope; no cross-lingual modeling for Vietnamese–English mixing

A. MONOLINGUAL TRANSFORMER ENCODERS

Early research on SMP detection primarily relied on rule-based frameworks designed for mobile environments [12]. Subsequently, recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) models [13], were adopted to learn sequential representations from message text. LSTM networks are effective at capturing local and mid-range dependencies, making them suitable for tasks where word order is important. However, their reliance on sequential computation limits scalability to long sequences and restricts efficient parallelization during training.

The introduction of the Transformer architecture [8] marked a fundamental shift in natural language processing by replacing recurrence with self-attention, enabling efficient modeling of global contextual dependencies. Building on this paradigm, pretrained language models such as BERT [9] and RoBERTa [14] have demonstrated strong performance across a wide range of NLP tasks. BERT is pretrained on large-scale corpora using masked language modeling and next sentence prediction objectives, enabling deep bidirectional contextual representation learning. Transformer-based approaches have also been applied to phishing and SMS classification, where models such as SM-Detector [1], [15] show improved robustness compared to traditional machine learning methods.

In the Vietnamese NLP domain, monolingual Transformer encoders have been developed to better capture language-specific characteristics. PhoBERT [11], pretrained on large-scale Vietnamese corpora, incorporates Vietnamese-specific word segmentation [16] and subword modeling techniques such as SentencePiece [17] and byte-pair encoding (BPE) [18]. These design choices enable more coherent modeling of compound words, diacritics, and tonal morphology. As a result, PhoBERT achieves strong performance on tasks such as part-of-speech tagging, dependency parsing, and sentiment analysis. Related studies [19], [20], [21] further demonstrate the effectiveness of Transformer-based and cross-lingual architectures for Vietnamese sequence labeling and social media analysis, particularly in handling noisy and informal text.

Recent studies [22], [23] have explored hybrid architectures that combine pretrained Transformer encoders with BiLSTM layers. In such designs, the Transformer serves as a contextual feature extractor, while the BiLSTM models sequential dependencies to reinforce order-sensitive patterns. Similarly, VIESpam [24] demonstrates the effectiveness of BiLSTM-based modeling for bilingual SMS spam classification. However, these approaches rely primarily on sequential modeling and do not explicitly address compound-level obfuscation or tokenization inconsistencies in code-mixed inputs.

To address structural obfuscation, recent work has introduced compound-aware modeling strategies for smishing detection. For instance, Nguyen-Xuan et al. [25] propose an encoder-only Transformer framework that reconstructs semantic components from obfuscated compound tokens prior to encoding, improving the visibility of latent phishing cues. Table 1 summarizes the characteristics of representative Transformer-based encoder models in the context of SMP detection, with a focus on their ability to handle code-mixed inputs and compound-token structures.

B. MULTILINGUAL TRANSFORMER ENCODERS

The multilingual capability of mBERT has been systematically analyzed by Pires et al. [10], who demonstrate that mBERT enables effective zero-shot cross-lingual transfer despite being pretrained solely on monolingual corpora. Architecturally, mBERT extends the original BERT framework by jointly training a single model on multilingual Wikipedia corpora using a shared subword vocabulary, thereby enabling cross-lingual transfer without explicit

TABLE 2. Structural characteristics of existing datasets for SMP detection.

Feature	SMSSpamCollection [33]	VNSocialComments [35]	ChongLuaDaoV2 [7]	Proposed Corpus (Ours)
Text Availability	Full	Full	None	Full
URL Presence	Very Low	Very Low	Full	Controlled
SMP Style (Informal / Conversational)	None	Strong	None	Strong
Phishing Indicator (Text-based)	Moderate	Limited	None	Strong
Phishing Indicator (URL-based)	Limited	Rare	Strong	Strong
English Keyword Usage	Moderate	Low	High	High
Vietnamese Language Usage	None	High	Very Low	High
code-mixed (VI–EN)	None	Moderate	Present (URL)	Systematic
Code-Mix Structure	Not applicable	Not systematic	Not systematic	Systematic
Compound Token Presence	Low	Limited	High	Controlled
Compound Structure Type	None	Weak	Strong	Strong
Delimiter Usage (−, _)	Rare	Low	Moderate	High
Obfuscation Patterns	Low	Low	High	High
URL Path Semantics	Rare	Rare	Present	Present
Text–URL Interaction	None	None	None	Explicit
Scenario Diversity (SMP context)	Low	Moderate	Low	High (S1–S4)

alignment constraints. Building upon this paradigm, XLM-RoBERTa [26] improves multilingual representation learning by leveraging larger and more diverse training data, resulting in more robust cross-lingual representations and consistently stronger performance across multilingual benchmarks.

Subsequent studies have further explored multilingual pretraining strategies to enhance cross-lingual alignment. InfoXLM [27] introduces contrastive learning objectives to improve cross-lingual semantic alignment, while mT5 [28] extends the text-to-text Transformer framework to a massively multilingual setting, supporting a wide range of generation and understanding tasks. Similarly, RemBERT [29] improves parameter efficiency and representation quality for multilingual models by decoupling embedding and encoder scaling. These approaches demonstrate that large-scale multilingual pretraining is effective in capturing language-invariant semantic representations.

Language-agnostic BERT Sentence Embedding (LaBSE) is proposed by Feng et al. [30] as a multilingual sentence encoder capable of producing language-independent representations across more than 100 languages. Built upon a dual-encoder architecture, LaBSE is pretrained using a combination of masked language modeling and translation ranking objectives on large-scale parallel and monolingual corpora. This training strategy explicitly aligns semantically equivalent sentences within a shared embedding space, enabling strong performance in cross-lingual sentence similarity, retrieval, and classification tasks.

Despite their strong performance on standard benchmarks, multilingual Transformer encoders exhibit complementary limitations when applied to noisy, domain-specific, and mixed-language text. Models such as mBERT and XLM-RoBERTa rely on shared subword tokenization strategies [13], [30], which may dilute language-specific cues under compound and code-mixed conditions. In contrast, sentence-level embedding approaches such as Sentence-BERT [32] and LaBSE emphasize semantic alignment but may overlook fine-grained lexical interactions. These limitations become particularly pronounced in SMP scenarios involving informal writing, intra-token code-mixed, and structural obfuscation.

In general, monolingual and multilingual encoders remain limited in their ability to model code-mixed inputs and compound-token structures, which are common in SMP scenarios where Vietnamese and English tokens co-occur within the same message or even within the same token. While cross-lingual models have shown effectiveness in Vietnamese social media analysis [21], they do not explicitly address compound-level obfuscation and intra-token code-mixed, which are central challenges in SMP detection. Motivated by these observations, we propose a mixed-language fusion architecture that integrates language-specific and multilingual representations, enabling the model to jointly capture fine-grained token-level structure and cross-lingual semantic consistency.

C. VIETNAMESE-ENGLISH SMP RESOURCES

Early benchmarks such as the SMSSpamCollection dataset [33] provide labeled SMS messages with spam and phishing intent. However, they are limited to English text, contain minimal URL information, and do not reflect modern SMP environments where attacks often combine conversational content with embedded links. As a result, these datasets are insufficient for modeling multilingual and cross-modal phishing behaviors. Recent surveys [34] highlight that evolving spam strategies increasingly exploit linguistic obfuscation and domain shifts, making traditional detection methods less effective.

To better capture Vietnamese language usage, social media datasets such as the Vietnamese Social Comments [35] have been introduced. These datasets reflect realistic informal language, including slang, abbreviations, and partial code-mixed. However, they exhibit extremely low URL presence and lack explicit phishing intent or adversarial obfuscation patterns. Consequently, they are more suitable for general spam detection or language modeling tasks rather than phishing detection.

On the other hand, URL-focused datasets such as ChongLuaDaoV2 [7] provide large-scale collections of phishing and benign URLs with strong structural patterns, including domain obfuscation and compound-like token formations. While these datasets are effective for learning URL-level representations, they completely lack accompanying message text and therefore fail to model the interaction between textual context and malicious links, which is essential in SMP scenarios.

Existing datasets are inherently modality-isolated, focusing either on textual content or URL structures, but rarely both within SMP contexts. This separation limits their ability to capture the interaction between linguistic cues and link-based deception, which is central to real-world smishing attacks. Consequently, current benchmarks fail to adequately reflect the structural complexity of modern SMP phishing scenarios. Table 2 further substantiates these limitations by comparing the structural characteristics of representative datasets.

Moreover, critical properties such as compound-level obfuscation, systematic code-mixed, and SMP-style interaction are either weakly represented or entirely absent. Collectively, these shortcomings reveal a fundamental structural gap in existing resources and motivate the development of a unified, scenario-controlled dataset that jointly models linguistic, structural, and adversarial characteristics of SMP phishing.

III. MIXED-LANGUAGE TRANSFORMER ENCODER ARCHITECTURE

A. PROBLEM DEFINITION

Given an input social media message x, we denote its tokenized form as $X = \{x_1, x_2, \ldots, x_N\}$, where N is the number of tokens in the message. The objective of SMP detection is to determine whether X conveys malicious intent or benign content. Formally, the task is to learn a mapping

$$f : X \rightarrow \{0, 1\}, \tag{1}$$

where label 1 corresponds to an SMP message and 0 to a non-SMP message.

A classifier produces an estimated probability $\hat{y} \in (0, 1)$ representing the likelihood that the message belongs to the SMP class. The prediction is obtained by comparing $\hat{y}$ against a decision threshold $\tau^{\star}$.

The central challenge lies in accurately modelling the token sequence X under realistic SMP conditions, where messages exhibit complex linguistic and structural characteristics. In particular, Vietnamese SMP messages frequently involve (i) Vietnamese–English code-mixed, (ii) intra-token compound formations, and (iii) adversarial obfuscation patterns. These phenomena lead to irregular token boundaries and semantic fragmentation, which are not effectively handled by standard tokenisation and encoding schemes.

Moreover, phishing attacks in SMP often rely on implicit intent expressed through informal conversational language rather than explicit keywords, requiring models to capture both local lexical cues and global contextual dependencies within noisy and adversarial environments.

Algorithm 1 Code-Mixed Decomposition

Input: Raw message x
Output: Normalized message x'
`// Step 1: Canonical normalization`
$x \leftarrow \text{NFC}(x)$
$x \leftarrow \text{ConfusableNorm}(x)$
$x \leftarrow \text{Lowercase}(x)$
$x \leftarrow \text{WhitespaceNorm}(x)$
`// Step 2: Light punctuation cleaning`
$x \leftarrow \text{RemoveDecorativeSymbols}(x)$
`// Step 3: Light decomposition`
foreach *token w in x* **do**
  $w \leftarrow \text{SplitAtCharClassTransition}(w)$
`// Step 4: Preserve code-mixed structures`
$x' \leftarrow \text{PreserveMixedCode}(x)$
return x'

Figure 1 illustrates the proposed MLTEA for SMP detection. The architecture processes each input message through six sequential stages: (1) a preprocessing layer that normalises surface variation while preserving code-mixed structure, (2) parallel PhoBERT and XLM-RoBERTa encoders that produce complementary token-level representations, (3) an offset-based alignment step that maps encoder-specific subword representations to a shared reference token grid, (4) a token-level fusion layer that integrates the aligned representations via a learnable projection, (5) a BiLSTM sequence encoder that reinforces local sequential dependencies, and (6) a gated pooling and classification head that produces the final SMP or non-SMP decision. Each component is described in detail in the following subsections.

B. PREPROCESSING LAYER

The preprocessing layer comprises two stages: (i) code-mixed decomposition and (ii) subword tokenisation. Rather than performing aggressive text rewriting, the strategy is designed to stabilise surface-form statistics by correcting adversarial spelling variations and informal orthographic distortions, while preserving the original lexical and semantic structure of the input. This conservative design prevents the introduction of artificial distributional shifts between the processed text and the representations learned by pretrained language models.

Algorithm 1 normalizes raw social media messages while preserving linguistically and semantically informative structures. Its primary objective is to reduce superficial variation and intentional obfuscation without over-fragmenting tokens that encode meaningful code-mixed or alphanumeric patterns. The decomposition process in Algorithm 1 consists of four key stages:

- Step 1 (Canonical normalization): Standardizes encoding, mitigates homoglyph obfuscation, and removes case and spacing variation.

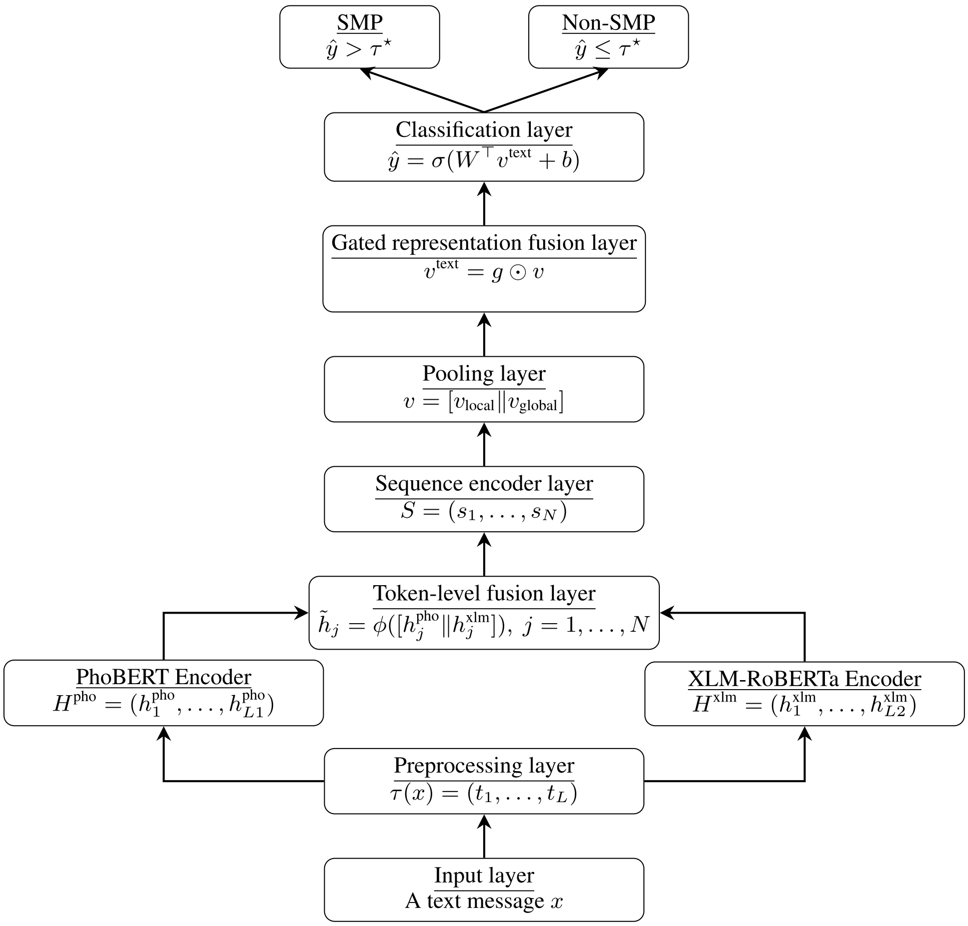

FIGURE 1. **Proposed MLTEA for SMP detection.**

- Step 2 (Light punctuation cleaning): Removes non-informative symbols while retaining structurally meaningful delimiters (e.g., ., /, −, _) commonly observed in URLs and identifiers.
- Step 3 (Light decomposition): Applies token-level splitting at character-class transitions (letter–digit, letter–symbol), enabling partial decomposition of fused alphanumeric tokens without disrupting coherent structures such as URLs.
- Step 4 (Preservation of code-mixed structures): Retains semantically critical elements, including URLs and code-mixed tokens.

Algorithm 2 performs subword tokenisation, mapping the normalized text x' into fixed-length numerical representations for Transformer-based encoders. Given a tokenizer $\tau(\cdot)$ and maximum sequence length T, it produces a token-ID sequence $\mathbf{x} \in \mathbb{N}^T$ and an attention mask $\mathbf{m} \in \{0, 1\}^T$. If the tokenized length $L < T$, the sequence is right-padded with corresponding mask values set to 0. Otherwise, if $L > T$, it is truncated to the first T tokens.

Algorithm 2 Subword Tokenisation

Input: Normalised message x', tokeniser $\tau(\cdot)$, maximum length T
Output: Token-ID sequence $\mathbf{x} \in \mathbb{N}^T$, attention mask $\mathbf{m} \in \{0, 1\}^T$

```
// Step 1: Subword segmentation
```
$(t_1, \ldots, t_L) \leftarrow \tau(x')$
```
// Step 2: Sequence length handling
```
if $L < T$ **then**
  Pad $(t_1, \ldots, t_L)$ to length T
  $\mathbf{m}_i \leftarrow 1$ for $i \leq L$
  $\mathbf{m}_i \leftarrow 0$ for $i > L$
else
  Truncate $(t_1, \ldots, t_L)$ to length T
  $\mathbf{m}_i \leftarrow 1$ for all $i \leq T$
```
// Step 3: Token-ID mapping
```
$\mathbf{x} \leftarrow \text{TokenID}(t_1, \ldots, t_T)$
return $\mathbf{x}, \mathbf{m}$

C. DUAL ENCODERS AND OFFSET-BASED ALIGNMENT

The normalized message x' is processed in parallel by two pretrained Transformer encoders with complementary linguistic strengths. PhoBERT [11], pretrained on large-scale Vietnamese corpora with language-specific word segmentation and SentencePiece byte-pair encoding, produces token-level representations

$$H^{\text{pho}} = \left(h_1^{\text{pho}},\, h_2^{\text{pho}},\, \ldots,\, h_{L_1}^{\text{pho}}\right), \tag{2}$$

that capture Vietnamese morphology, compound word structure, and informal writing patterns.

XLM-RoBERTa [26], pretrained on large-scale Common-Crawl data spanning 100 languages, generates multilingual token-level representations

$$H^{\text{xlm}} = \left(h_1^{\text{xlm}},\, h_2^{\text{xlm}},\, \ldots,\, h_{L_2}^{\text{xlm}}\right), \tag{3}$$

that encode cross-lingual contextual information and English lexical cues frequently exploited in phishing content.

Because PhoBERT and XLM-RoBERTa employ distinct tokenizers and vocabularies, they typically produce sequences of different lengths ($L_1 \neq L_2$) for the same input. To reconcile this discrepancy, we adopt an offset-based alignment mechanism rather than enforcing a shared tokenization scheme, thereby preserving the representational strengths of each pretrained encoder.

Specifically, the normalized message x' is segmented into a sequence of reference tokens $\{w_j\}_{j=1}^{N}$ using a whitespace- and delimiter-aware preprocessing step. Each reference token w_j is associated with a character interval $\mathcal{I}(w_j)$ in the original text. A subword token t_i from encoder e is aligned to w_j if and only if

$$\text{span}(t_i) \cap \mathcal{I}(w_j) \neq \emptyset. \tag{4}$$

As each reference token may correspond to a variable number of subword tokens, we apply mean pooling over all aligned subword representations to obtain a unified token-level representation at the reference-token granularity:

$$h_j^{(e)} = \text{MeanPool}\Big(\big\{h_i^{(e)} \mid \text{span}(t_i^{(e)}) \cap \mathcal{I}(w_j) \neq \emptyset\big\}\Big),\quad e \in \{\text{pho},\ \text{xlm}\} \tag{5}$$

Algorithm 3 summarises the complete procedure. After alignment, both encoders yield representations at the same N-token granularity, enabling direct token-level fusion.

Algorithm 3 Offset-Based Cross-Encoder Alignment

Input: Normalised message x', reference token spans $\{\mathcal{I}(w_j)\}_{j=1}^{N}$, tokenisers τ_{pho}, τ_{xlm}

Output: Aligned representations $\{h_j^{\text{pho}}\}_{j=1}^{N}$, $\{h_j^{\text{xlm}}\}_{j=1}^{N}$

`// Step 1: Encoder-specific tokenisation`

$(t_1^{\text{pho}}, \ldots, t_{L_1}^{\text{pho}}) \leftarrow \tau_{\text{pho}}(x')$

$(t_1^{\text{xlm}}, \ldots, t_{L_2}^{\text{xlm}}) \leftarrow \tau_{\text{xlm}}(x')$

Obtain character spans $\text{span}(t_i^{\text{pho}})$ and $\text{span}(t_i^{\text{xlm}})$ for all i

`// Step 2: Encode subword representations`

$\{h_i^{\text{pho}}\}_{i=1}^{L_1} \leftarrow \text{PhoBERT}(t_1^{\text{pho}}, \ldots, t_{L_1}^{\text{pho}})$

$\{h_i^{\text{xlm}}\}_{i=1}^{L_2} \leftarrow \text{XLM-R}(t_1^{\text{xlm}}, \ldots, t_{L_2}^{\text{xlm}})$

`// Step 3: Offset-based alignment and pooling`

for $j \leftarrow 1$ **to** N **do**

  $\mathcal{S}_j^{\text{pho}} \leftarrow \{i \mid \text{span}(t_i^{\text{pho}}) \cap \mathcal{I}(w_j) \neq \emptyset\}$

  $\mathcal{S}_j^{\text{xlm}} \leftarrow \{i \mid \text{span}(t_i^{\text{xlm}}) \cap \mathcal{I}(w_j) \neq \emptyset\}$

  $h_j^{\text{pho}} \leftarrow \text{MeanPool}(\{h_i^{\text{pho}} \mid i \in \mathcal{S}_j^{\text{pho}}\})$

  $h_j^{\text{xlm}} \leftarrow \text{MeanPool}(\{h_i^{\text{xlm}} \mid i \in \mathcal{S}_j^{\text{xlm}}\})$

return $\{h_j^{\text{pho}}\}_{j=1}^{N}$, $\{h_j^{\text{xlm}}\}_{j=1}^{N}$

D. TOKEN-LEVEL FUSION LAYER

After alignment, both encoders yield representations at the same reference-token granularity. To integrate these complementary signals, token-level fusion is performed at each position j:

$$\tilde{h}_j = \phi\big([h_j^{\text{pho}} \,\|\, h_j^{\text{xlm}}]\big), \quad j = 1, \ldots, N, \tag{6}$$

where $[\cdot \,\|\, \cdot]$ denotes vector concatenation and $\phi(\cdot) = \text{ReLU}(W_\phi \cdot + b_\phi)$ is a position-wise linear projection with learnable parameters $W_\phi \in \mathbb{R}^{d \times 2d}$ and $b_\phi \in \mathbb{R}^{d}$, mapping the concatenated $2d$-dimensional vector back to the original hidden dimension d.

Performing fusion at the token level rather than at the sequence level preserves fine-grained bilingual information and enables explicit modelling of code-mixed compound tokens before any sequence-level aggregation occurs.

E. BILSTM SEQUENCE ENCODER

The fused token sequence $(\tilde{h}_1, \ldots, \tilde{h}_N)$ is passed to a two-layer Bidirectional Long Short-Term Memory (BiLSTM) network [13], [24] to model sequential dependencies:

$$S = (s_1, s_2, \ldots, s_N). \tag{7}$$

Each BiLSTM layer uses a hidden size of 256 per direction (total output dimension 512), with dropout of 0.2 applied between layers. While Transformer self-attention captures global contextual relationships, the BiLSTM explicitly models local ordering and directional dependencies [13], [24], which are particularly important for detecting highly obfuscated phishing patterns in code-mixed text where token order encodes intent.

F. POOLING LAYER

To obtain a fixed-dimensional representation suitable for classification, the BiLSTM output sequence S is aggregated via two complementary pooling operations:

$$v = [v_{\text{local}} \,\|\, v_{\text{global}}], \tag{8}$$

where $v_{\text{local}} = \text{MaxPool}(S)$ and $v_{\text{global}} = \text{AvgPool}(S)$.

Max-pooling selects the most salient phishing indicator at each feature dimension, making it effective for detecting sparse but discriminative cues such as credential-related keywords embedded within code-mixed tokens. Average-pooling captures the overall semantic context of the message, providing a complementary global summary that is less sensitive to individual token activations. Concatenating both pooling outputs allows the classifier to jointly exploit local salience and global context, a strategy empirically validated for text classification tasks [24]. Alternative pooling strategies, including `[CLS]` token-only and attention-weighted pooling, were evaluated during preliminary experiments. The max–average concatenation consistently achieved the best validation F_1 score and was therefore adopted.

G. GATED REPRESENTATION FUSION

After pooling, a dynamic input-conditioned gating mechanism is applied to adaptively reweight feature dimensions based on the characteristics of each input message:

$$v^{\text{text}} = g \odot v, \tag{9}$$

where $g = \sigma(W_g v + b_g)$, $W_g \in \mathbb{R}^{2d \times 2d}$, $b_g \in \mathbb{R}^{2d}$ are learnable parameters, $\sigma(\cdot)$ is the sigmoid activation, and $\odot$ denotes element-wise multiplication.

Critically, g is *computed from the input* v rather than being a static learned mask, making it a true input-dependent gating mechanism. For a given message, dimensions corresponding to language-specific features (e.g., Vietnamese morphological cues from PhoBERT) or cross-lingual features (e.g., English credential keywords from XLM-RoBERTa) are dynamically up-weighted or suppressed according to the message's linguistic composition. This adaptive weighting is particularly beneficial under structurally perturbed conditions, where the relative informativeness of monolingual versus multilingual features varies across messages. It also improves prediction score calibration, producing sharper class separation that enables more effective threshold optimisation (Section III-I).

H. CLASSIFICATION LAYER

The gated representation v^{text} is passed to a sigmoid-based binary classifier:

$$\hat{y} = \sigma(W^{\top} v^{\text{text}} + b), \tag{10}$$

which outputs the probability that the input message is SMP. A decision threshold $\tau^{\star}$, optimised on the validation set, is applied to produce the final binary label.

I. OPTIMAL DECISION THRESHOLD

The F_1 score is the harmonic mean of precision and recall. For a given threshold τ, predicted probabilities are binarised as

$$\hat{y}^{(\tau)} = \mathbf{1}[\hat{y} > \tau], \tag{11}$$

from which true positives (TP), false positives (FP), and false negatives (FN) are computed. Precision, recall, and the resulting F_1 score are:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$
$$F_1 = \frac{2TP}{2TP + FP + FN}. \tag{12}$$

Algorithm 4 Optimal Decision Threshold Search

Input: Validation set $\mathcal{D}_{\text{val}}$, predicted scores $\{\hat{y}_i\}$, candidate thresholds $\mathcal{T} \subset [0, 1]$
Output: Optimal threshold $\tau^{\star}$

$F_1^{\text{best}} \leftarrow 0$
$\tau^{\star} \leftarrow 0.5$
foreach $\tau \in \mathcal{T}$ **do**
  Compute labels $\hat{y}_i^{(\tau)} \leftarrow \mathbf{1}[\hat{y}_i > \tau]$
  Compute TP, FP, FN over $\mathcal{D}_{\text{val}}$
  $F_1(\tau) \leftarrow \dfrac{2TP}{2TP + FP + FN}$
  if $F_1(\tau) > F_1^{\text{best}}$ **then**
    $F_1^{\text{best}} \leftarrow F_1(\tau)$
    $\tau^{\star} \leftarrow \tau$
return $\tau^{\star}$

Because neural classifiers particularly those involving fused or mixed-language representations often produce uncalibrated probability estimates, the default threshold $\tau = 0.5$ may not correspond to the optimal decision boundary. To identify the most effective operating point, we evaluate $F_1(\tau)$ over a dense grid of candidate thresholds on the validation set and select

$$\tau^{\star} = \arg \max_{\tau \in [0,1]} F_1(\tau). \tag{13}$$

For in-distribution evaluation on D_0, the optimal threshold $\tau^{\star}$ is determined using the validation split of D_0, and for cross-dataset robustness evaluation on D_{15} and D_{30}, $\tau^{\star}$ is selected solely from this validation split and then held fixed during evaluation on D_{15} and D_{30}. This protocol prevents any structural information from the perturbed datasets from influencing threshold selection, thereby ensuring an unbiased assessment of robustness, as summarised in Algorithm 4.

IV. EXPERIMENTAL DESIGN

A. DATASET CONSTRUCTION AND SPLITTING

To evaluate mixed-language and compound-level robustness in SMP, we construct a scenario-controlled corpus grounded in real-world datasets and augmented via a structural obfuscation operator.

1) SOURCE DATASETS AND SCENARIO MAPPING (S1–S4)

The corpus is derived from three publicly available datasets: the Vietnamese Social Comments dataset [35], the SMSSpamCollection dataset [33], and the ChongLuaDaoV2 dataset [7]. These datasets are mapped into four semantic–stylistic scenarios:

- S1 (Benign Code-Mixed Interaction): conversational and informal Vietnamese content with optional code-mixed, sampled from Vietnamese Social Comments [35].

- S2 (Promotional/Spam Content): adversarial or marketing-oriented messages derived from the spam class of SMSSpamCollection [33].
- S3 (Legitimate Social Content): neutral and non-adversarial Vietnamese messages from Vietnamese Social Comments [35].
- S4 (Phishing-Oriented Content): URL-centric samples constructed from ChongLuaDaoV2 [7], optionally combined with templated message prompts to simulate realistic SMP contexts.

These scenarios capture complementary SMP dimensions, including lexical code-mixed, adversarial style, conversational noise, and phishing intent. Multiple characteristics may co-occur within a single message.

2) STRUCTURAL OBFUSCATION OPERATOR (S_5)

S_5 is a deterministic transformation $S_5 : \mathcal{X} \rightarrow \mathcal{X}$ that preserves semantic intent while modifying token structure. It models realistic obfuscation patterns observed in phishing URLs and messages, including delimiter-based fusion (e.g., "ma_otp"), compound-domain construction (e.g., "ma_otp.bank.vn"), homograph substitution, and alphanumeric perturbation.

3) CORPUS CONSTRUCTION

We construct a corpus of 10,000 messages via balanced sampling from the three source datasets, comprising 5,000 SMP and 5,000 non-SMP instances. Each sample is assigned a scenario label (S1–S4) according to its source and semantic intent, and may be further transformed using S_5 to introduce controlled structural variation. This process preserves the underlying linguistic characteristics of real-world data while enabling systematic manipulation of obfuscation patterns.

4) CONTROLLED STRUCTURAL SETTINGS

To enable systematic evaluation of structural robustness, we derive three controlled evaluation settings D_0, D_{15}, and D_{30} with progressively increasing compound density. Compound density is defined at the token level as

$$\rho(D) = \frac{\sum_{m \in D} |\mathcal{F}(m)|}{\sum_{m \in D} L_m}, \tag{14}$$

where $\mathcal{F}(m)$ denotes structurally fused tokens and L_m denotes the token length of message m. The datasets are constructed such that compound density is approximately 0%, 15%, and 30% for D_0, D_{15}, and D_{30}, respectively.

These settings simulate progressively stronger structural obfuscation while preserving semantic intent and class balance. Other corpus properties, including code-mix rate and URL presence, are not explicitly controlled but remain within a narrow range due to the underlying source distributions. This design allows compound density to serve as the primary structural variable while acknowledging minor residual variation across settings.

TABLE 3. Structural statistics of constructed SMP evaluation settings.

Dataset	Code-Mix Rate (%)	Compound Density (%)	URL Presence (%)
D_0	29.12	0	20.36
D_{15}	29.48	15	21.27
D_{30}	29.85	30	22.18

5) CODE-MIX CHARACTERIZATION

The code-mix rate is defined at the message level as

$$\gamma(D) = \frac{1}{|D|} \sum_{m \in D} \delta_{\text{cm}}(m), \tag{15}$$

where $\delta_{\text{cm}}(m) = 1$ if message m contains both Vietnamese and English tokens. The observed code-mix rate is primarily inherited from the Vietnamese Social Comments dataset and remains relatively stable across settings, reflecting natural bilingual variation rather than controlled manipulation.

6) DATA SPLITTING AND EVALUATION PROTOCOL

Each dataset is partitioned into 70% training, 15% validation (Dev), and 15% testing splits using stratified sampling to preserve class balance and structural characteristics. The hyperparameters and decision thresholds are determined exclusively on the validation set. To assess structural robustness, models are trained on D_0 and evaluated on D_{15} and D_{30}, enabling controlled measurement of generalization under increasing compound-level obfuscation.

7) CODE-MIX CONFOUND

While compound density $\rho(D)$ is explicitly controlled, the code-mix rate $\gamma(D)$ is not independently regulated and exhibits only minor variation across D_0, D_{15}, and D_{30} (Table 3). Similarly, URL presence, influenced by ChongLuaDaoV2 [7] samples and structural transformations, increases slightly with higher compound density. As a result, performance differences across datasets may reflect a combination of structural obfuscation and small shifts in bilingual and URL-related features. However, these variations are limited in magnitude compared to the controlled increase in compound density, suggesting that compound formation remains the dominant structural factor.

8) FEATURE SOURCE ATTRIBUTION

The code-mix rate is largely influenced by Vietnamese Social Comments, compound structures are introduced through controlled transformations and SMSSpamCollection-derived patterns, and URL presence is primarily contributed by ChongLuaDaoV2. This separation ensures that each structural dimension reflects realistic characteristics of its corresponding source while maintaining controlled experimental variation.

9) LIMITATIONS AND EXTERNAL VALIDITY

While the proposed dataset is grounded in real-world sources, the structural obfuscation patterns introduced via S_5 are

TABLE 4. Training configuration and hyperparameter settings.

Parameter	Value
Encoder Backbone	
PhoBERT$_{\text{base}}$	12 layers, 12 heads, $d = 768$
XLM-RoBERTa$_{\text{base}}$	12 layers, 12 heads, $d = 768$
BiLSTM Sequence Encoder	
Layers	2 (stacked, bidirectional)
Hidden size	256 per direction (512 total)
Inter-layer dropout	0.2
Classification Head	
Activation	Sigmoid (binary classifier)
Loss function	Binary cross-entropy
Optimisation	
Optimizer	AdamW
Encoder LR (PhoBERT, XLM-RoBERTa)	2×10^{-5}
Classifier / BiLSTM head LR	1×10^{-4}
Weight decay	0.01
Batch size	32
Max epochs	10
Gradient clipping	ℓ_2-norm ≤ 1.0
Early stopping	Patience $= 3$ (val. F_1)
Hardware and Reproducibility	
Framework	PyTorch 2.1 / CUDA 11.8
GPU	NVIDIA (16 GB VRAM)
Random seed	42

programmatically controlled and may not fully capture the diversity and unpredictability of real-world attacker behavior. In practice, adversaries employ evolving strategies that may extend beyond the transformation space defined by S_5. Therefore, the dataset should be interpreted as a controlled benchmark for evaluating structural robustness rather than a complete representation of all SMP attack patterns.

B. TRAINING CONFIGURATION

All models are trained under identical optimisation settings to ensure fair comparison. Following Section IV-A, training is conducted on D_0, while D_{15} and D_{30} are reserved for robustness evaluation.

MLTEA integrates pretrained encoders: PhoBERT$_{\text{base}}$ [11] and XLM-RoBERTa$_{\text{base}}$ [26], each with 12 layers, 12 attention heads, and hidden size $d = 768$. Both encoders are fully fine-tuned without modification. Single-encoder baselines are trained under identical settings for consistency.

The sequence encoder consists of a two-layer bidirectional BiLSTM with hidden size 256 per direction (512 output), with inter-layer dropout of 0.2. A sigmoid classification head outputs the SMP probability.

Optimisation uses AdamW [36] with weight decay 0.01, batch size 32, and up to 10 epochs. Gradient clipping ($\ell_2 \leq 1.0$) and early stopping (patience 3, based on validation F_1) are applied.

A differential learning rate schedule is used: 2×10^{-5} for pretrained encoders and 1×10^{-4} for the BiLSTM and classifier, enabling stable fine-tuning while accelerating convergence of randomly initialised components.

Experiments are implemented in PyTorch 2.1 with CUDA 11.8 on a single NVIDIA GPU (16 GB VRAM).

TABLE 5. Precision, recall, and F_1 score on D_0 (test split).

Model	Precision	Recall	F_1 score
BiLSTM	0.88	0.87	0.87
PhoBERT$_{\text{base}}$	0.89	0.87	0.88
RoBERTa$_{\text{base}}$	0.88	0.86	0.87
XLM-RoBERTa$_{\text{base}}$	0.88	0.86	0.87
PhoBERT$_{\text{base}}$ + BiLSTM	0.92	0.90	0.91
XLM-RoBERTa$_{\text{base}}$ + BiLSTM	0.93	0.90	0.91
MLTEA (proposed)	0.94	0.94	0.94

Random seeds are fixed to 42. Hyperparameters are summarised in Table 4.

C. BASELINE MODELS

To evaluate the effectiveness of the proposed MLTEA, we compare it against six representative baselines spanning recurrent, monolingual, multilingual, and hybrid architectures. All baseline models are trained on the training split of D_0 and evaluated using the same data splits and optimisation settings described in Section IV-A, ensuring a fair comparison under the controlled structural evaluation protocol.

To maintain consistency across models, we apply an identical decision-threshold optimisation procedure (Section III-I), where the optimal threshold $\tau^\star$ is selected based on the validation split of D_0.

- BiLSTM: A standard recurrent classifier using 100-dimensional GloVe embeddings [37]. A two-layer bidirectional LSTM (256 hidden units per direction) encodes the sequence, with max pooling applied over hidden states before a sigmoid classifier. This model represents traditional sequence modelling without contextual pretraining and serves as a lower-bound baseline.
- PhoBERT$_{\text{base}}$: A Vietnamese monolingual Transformer encoder fine-tuned with a linear classification head over the `[CLS]` representation. It captures rich Vietnamese morphology and compound-word structure, serving as a strong monolingual baseline.
- RoBERTa$_{\text{base}}$: A general-purpose English Transformer encoder fine-tuned for binary SMP classification. This baseline evaluates performance under English-centric representations, isolating the contribution of English-dominant lexical cues in code-mixed messages.
- XLM-RoBERTa$_{\text{base}}$: A multilingual Transformer encoder pretrained on 100 languages and fine-tuned for binary SMP classification. This baseline serves as a single-encoder multilingual reference, isolating the effect of incorporating a monolingual Vietnamese encoder.
- PhoBERT$_{\text{base}}$ + BiLSTM: A hybrid model where PhoBERT contextual embeddings are processed by a two-layer BiLSTM (256 hidden units per direction, dropout 0.2). The outputs are aggregated via max and average pooling, concatenated, and fed to a binary classifier. This baseline evaluates whether sequential modelling enhances monolingual contextual representations for SMP detection.

TABLE 6. F_1 score under progressive compound-level obfuscation.

Model	D_0	D_{15}	D_{30}	Δ_{30}
BiLSTM	0.87	0.81	0.76	-0.11
PhoBERT$_{\text{base}}$	0.88	0.83	0.79	-0.09
RoBERTa$_{\text{base}}$	0.87	0.82	0.77	-0.10
XLM-RoBERTa$_{\text{base}}$	0.87	0.84	0.80	-0.07
PhoBERT$_{\text{base}}$ + BiLSTM	0.91	0.87	0.83	-0.08
XLM-RoBERTa$_{\text{base}}$ + BiLSTM	0.91	0.87	0.84	-0.07
MLTEA (proposed)	0.94	0.93	0.92	-0.02

- XLM-RoBERTa$_{\text{base}}$ + BiLSTM: A hybrid model using XLM-RoBERTa as the multilingual encoder, followed by a two-layer BiLSTM with the same configuration as above. This baseline isolates the effect of dual-encoder fusion in MLTEA by replacing parallel PhoBERT and XLM-RoBERTa encoders with a single multilingual encoder while keeping the downstream architecture identical.

V. EXPERIMENTAL RESULTS

A. IN-DISTRIBUTION PERFORMANCE ON D_0

Table 5 reports precision, recall, and F_1 score on D_0, with $\tau^\star$ selected on the validation split and applied consistently. MLTEA achieves the best performance (0.94 across all metrics), outperforming all baselines.

Single-encoder Transformers (PhoBERT, RoBERTa, XLM-RoBERTa) cluster at $F_1 = 0.87$, indicating that neither monolingual nor multilingual modelling alone sufficiently captures compound-aware, code-mixed SMP characteristics. Notably, RoBERTa matches XLM-RoBERTa (both 0.87), suggesting limited benefit from multilingual coverage without explicit compound modelling.

Hybrid models (Transformer + BiLSTM) improve performance by approximately 0.03 F_1, confirming that sequential modelling provides complementary inductive bias for structurally obfuscated inputs. MLTEA further outperforms the strongest hybrid baseline (0.91) by 0.03, demonstrating that the gain arises from token-level dual-encoder fusion and gated reweighting rather than model capacity alone.

The evaluated baselines represent key architectural families, BiLSTM, single-encoder Transformers (monolingual/multilingual), and hybrid Transformer + BiLSTM models, ensuring controlled comparison under identical data splits, optimisation settings, and threshold selection.

B. STRUCTURAL ROBUSTNESS ANALYSIS

To evaluate robustness under compound-level obfuscation, all models are trained on the training split of D_0 and evaluated on the test splits of D_{15} and D_{30}, where compound density $\rho(D)$ increases progressively. This protocol isolates the impact of structural perturbation on model generalisation. For consistency, the decision threshold $\tau^\star$ is selected on the validation split of D_0 and fixed across all cross-dataset evaluations (Section IV-A). Table 6 reports the resulting F_1 scores. The results in Table 6 reveal three consistent patterns under increasing structural perturbation.

TABLE 7. Ablation study: F_1 score under controlled structural perturbation.

Model	D_0	D_{15}	D_{30}	Δ_{30}
Single-Backbone (XLM-RoBERTa$_{\text{base}}$)	0.91	0.86	0.82	-0.09
Dual-Backbone + Concatenation	0.92	0.89	0.86	-0.06
MLTEA (proposed)	0.94	0.93	0.92	-0.02

1) UNIVERSAL DEGRADATION UNDER OBFUSCATION

All baseline models exhibit monotonic declines in F_1 score as compound density increases, indicating that delimiter-based fusion, compound-token construction, and alphanumeric perturbations disrupt standard subword tokenisation and contextual encoding. The degradation is most pronounced for single-encoder models (Δ_{30} of -0.08 to -0.11), which rely solely on monolingual or multilingual representations and lack mechanisms to recover disrupted compound-token structure.

2) PARTIAL MITIGATION FROM SEQUENTIAL MODELING

Hybrid architectures (Transformer + BiLSTM) consistently outperform their single-encoder counterparts across all settings and show slightly reduced degradation (Δ_{30} of -0.07 to -0.08). This suggests that sequential modelling provides complementary inductive bias for handling local structural distortions. However, performance still declines with increasing $\rho(D)$, indicating that sequential modelling alone is insufficient to fully recover semantic coherence from fragmented compound tokens.

3) EXPLICIT COMPOUND-LEVEL ENCODING IMPROVES ROBUSTNESS

MLTEA achieves the highest F_1 scores across all settings and exhibits a substantially smaller performance drop ($\Delta_{30} = -0.02$) than all baselines. This near-flat degradation indicates that the combination of language-specific encoders and token-level fusion better preserves compound-token representations under increasing obfuscation. Unlike hybrid models that apply sequential modelling post hoc, MLTEA captures cross-lingual compound structure directly at the representation level, enabling more robust recovery of semantic coherence. These results demonstrate that robustness to structural obfuscation in code-mixed phishing detection requires explicit modelling of compound-level cross-lingual interactions, rather than relying solely on post hoc sequential compensation.

C. ABLATION STUDY

To quantify the contribution of each architectural component, we conduct an ablation study under the same controlled structural evaluation protocol. All variants are trained on the training split of D_0 and evaluated on D_0, D_{15}, and D_{30} to assess performance under clean, intermediate, and heavily perturbed conditions. We compare three variants:

1) Single-Backbone (XLM-RoBERTa$_{\text{base}}$): A single multilingual encoder followed by a BiLSTM and

TABLE 8. Effect of threshold optimisation on F_1 score (%).

Fusion Strategy	$F_1(\tau{=}0.5)$	$F_1(\tau^\star)$	ΔF_1
Non-gated fusion	93.42%	93.81%	+0.39
Gated fusion (MLTEA)	93.60%	94.47%	+0.87

classification head identical to MLTEA, isolating the contribution of dual-encoder fusion.

2) Dual-Backbone + Concatenation: PhoBERT and XLM-RoBERTa representations concatenated at the token level without adaptive weighting, followed by the same BiLSTM and classification head. This variant isolates the contribution of the gating mechanism.
3) Dual-Backbone + Gated Fusion (MLTEA): The full proposed architecture with input-conditioned learnable gating (Section III-G).

For each variant, the decision threshold $\tau^\star$ is selected on the validation split of D_0 and applied consistently across all settings. MLTEA scores are reported under independent threshold optimisation for each variant and coincide with those in Table 6 to two decimal places.

Table 7 reports F_1 scores across all settings. The single-backbone variant achieves 0.91 on D_0 but degrades to 0.82 on D_{30} ($\Delta_{30} = -0.09$), indicating limited robustness under compound-level obfuscation. The dual-backbone concatenation model improves both performance and robustness across all settings ($\Delta_{30} = -0.06$), demonstrating that combining monolingual and multilingual representations provides complementary cues; however, the absence of adaptive weighting limits the extent of this gain.

In contrast, MLTEA achieves the highest F_1 scores of 0.94, 0.93, and 0.92 on D_0, D_{15}, and D_{30}, respectively, with the smallest degradation ($\Delta_{30} = -0.02$), indicating that gated fusion effectively preserves compound-token representations under structural perturbation. These results confirm that both dual-encoder fusion and input-conditioned gating are essential for robust SMP detection.

D. EFFECT OF GATED FUSION ON DECISION CALIBRATION

As shown in Table 8, threshold optimisation improves performance for both models, with a substantially larger gain for gated fusion (+0.87 vs. +0.39). This suggests that the gated model produces more separable prediction scores around the decision boundary, making performance more responsive to threshold adjustment. The improvement is consistent with the role of adaptive gating, which enhances class-discriminative representations by selectively emphasising informative features under mixed-language and structurally obfuscated inputs.

1) CONFUSION MATRIX ANALYSIS

Figure 2 presents the normalised confusion matrix for MLTEA at $\tau^\star$ on the D_0 test split. The model exhibits balanced classification, with a false negative rate of 0.056 and a false positive rate of 0.054. This near-symmetric error

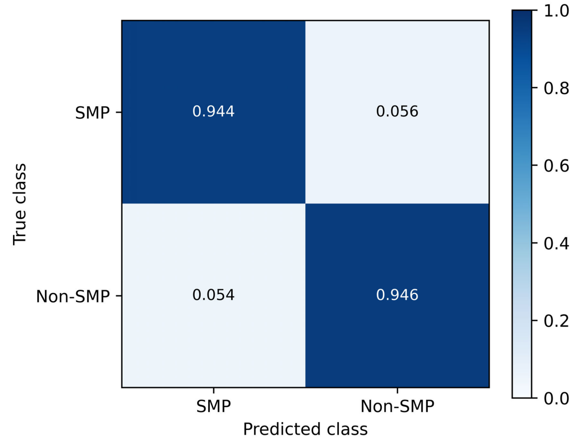

FIGURE 2. Normalised confusion matrix for MLTEA at $\tau^\star$ on the D_0 test split.

profile suggests that gated fusion combined with threshold optimisation produces a well-calibrated decision boundary, maintaining high recall without sacrificing precision under compound-rich, mixed-language conditions.

2) ERROR ANALYSIS

Misclassifications fall into three main categories. First, very short messages (e.g., *"ok nha"*, *"good nhé"*) provide insufficient semantic or structural evidence for reliable discrimination. Second, heavily code-mixed slang, emojis, and repeated characters introduce surface noise that degrades encoder representations. Third, a smaller set of errors arises from subtle promotional content embedded in otherwise benign conversational messages, where phishing intent is implicit. In these cases, the gating mechanism often assigns near-uniform weights across encoders, reducing representational contrast and leading to uncertain predictions. These observations motivate future work on uncertainty-aware prediction and character-level modelling.

VI. CONCLUSION AND FUTURE WORK

We propose MLTEA, a dual-encoder architecture that integrates monolingual Vietnamese (PhoBERT) and multilingual ($\text{XLM-RoBERTa}_{\text{base}}$) representations via offset-based token-level fusion and an input-conditioned gating mechanism. The model targets Vietnamese–English code-mixed compound structures, a key obfuscation strategy in SMP that standard Transformers struggle to capture due to subword fragmentation effects.

Experiments on a scenario-controlled corpus under progressively increasing compound density yield three main findings. First, MLTEA achieves an F_1 score of 0.94 on D_0, outperforming all baselines, with the strongest baseline (XLM-RoBERTa + BiLSTM) reaching 0.91. Second, MLTEA maintains strong performance under structural perturbation, achieving 0.92 on D_{30} with minimal degradation ($\Delta_{30} = -0.02$), compared to -0.07 to -0.11 for baselines.

Third, gated fusion and threshold optimisation provide complementary benefits, with larger gains from threshold tuning (+0.87 vs. +0.39 in F_1 score), indicating improved score separability and more stable decision boundaries.

The structural transformations are programmatically controlled and may not fully capture real-world attacker diversity. The code-mix rate $\gamma(D)$ is not independently controlled, so performance differences may partially reflect shifts in bilingual distribution. In addition, the current framework operates on text alone and does not model URL structure, limiting its ability to capture cross-modal phishing signals. Future directions include incorporating character-level or subword-aware representations to better handle heavily obfuscated tokens and adversarial perturbations. Extending MLTEA to jointly model message text and URL structure could further improve detection by capturing interactions between linguistic cues and link-based deception.

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